

PHYS 2801 - Assignment 2

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September 30, 2026

1 Question 1: The Maxwell–Boltzmann distribution

The Maxwell–Boltzmann distribution:

1.1 Part (a)

Here is the Maxwell–Boltzmann distribution function:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp \left(-\frac{mv^2}{2k_B T} \right) \quad (1)$$

The Maxwell-Boltzmann distribution tells us what fraction of the particles are moving slow or fast. If integrated over all speeds from 0 to infinity, the result is 1. f is the probability density, v is the particle speeds, m is the mass of the particles, T is the temperature, and k_B is the Boltzmann constant.

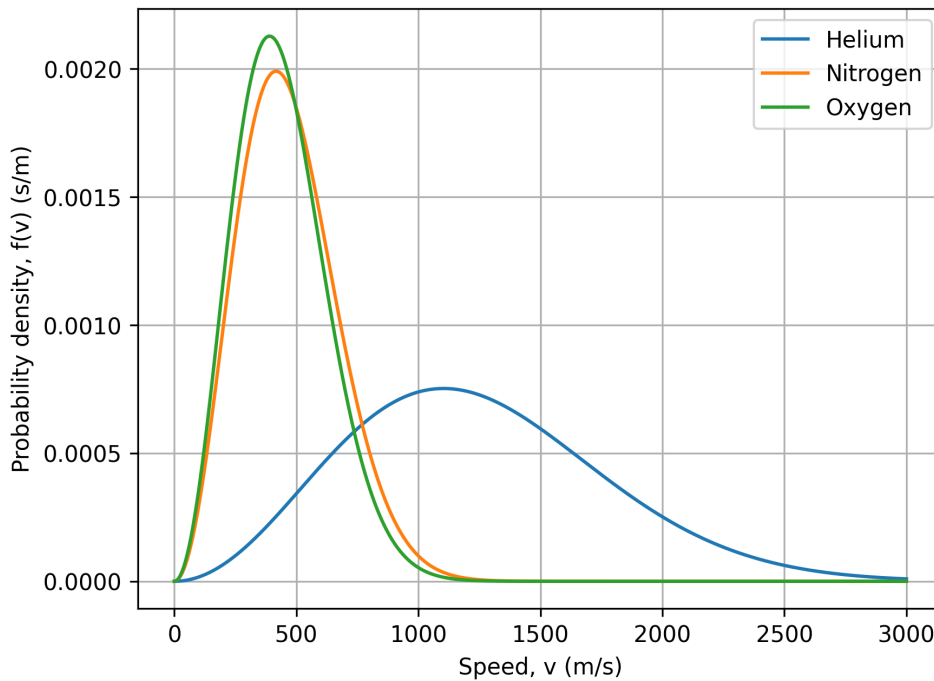


Figure 1: The Maxwell-Boltzmann distribution of particle speeds at temperature $T=293.15\text{K}$. Notice how Helium, because of being lighter has a distribution shifted more to the right meaning a bigger percentage of the helium has faster speeds compared to the other two elements. This plot was made using the equation as given in Eq. 1

c) cxx files are instructions for a computer on a higher end, meaning it is more readable like a human language. exe files are more low end meaning it is closer to how computers talk. Because of this, lower end exe files run much faster than the higher end counterparts because higher level languages have to be translated to machine code before running.

d) Based on the text output file I'd say the programmer was Indian, because the image looks like the map of India.